

ESC 425 Planetary Seismoacoustics

Course Project Report

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University of Rochester



Project Topic:

Improving Ambient Noise Correlations with Multi-Taper Method

Discuss Broadly Your Goals:

The objective of this project is to evaluate the effectiveness of the multi-taper signal processing technique in enhancing ambient noise cross-correlations for the extraction of surface waves, with a particular focus on Love waves. While Rayleigh waves typically exhibit stronger amplitudes and therefore higher signal-to-noise ratios (SNR) in cross-correlation results, Love waves often display greater stochasticity, leading to reduced correlation quality.

This study aims to demonstrate how the multi-taper approach can improve the quality and stability of cross-correlation results for Love waves. Quantitative comparisons will be made by reporting the SNR values for both standard and multi-taper processing methods. Continuous ambient noise data recorded at the network of stations in Madagascar (Olugboji & Siyu, 2022) will be used to compute cross-correlations, which will then be stacked according to inter-station distances to enable qualitative evaluation of the correlation wavefield, with and without multi-taper processing. Additionally, where feasible, the cross-correlation analysis will be extended following the methodology of Tkalcic et al. (2020) to compare and validate coda-correlation results obtained with and without the proposed multi-taper technique.

Provide References (1 or More):

- Olugboji, Tolulope, and Siyu Xue. "A Short-Period Surface-Wave Dispersion Dataset for Model Assessment of Africa's Crust: ADAMA." Seismological Society of America 93.3 (2022): 1943-1959.
- Karnik, Santhosh, Justin Romberg, and Mark A. Davenport. "Thomson's multitaper method revisited." IEEE Transactions on Information Theory 68.7 (2022): 4864-4891
- Tkalcic, H., Pham, T.S. and Wang, S., 2020. The Earth's coda correlation wavefield: Rise of the new paradigm and recent advances. Earth-Science Reviews, 208, p.103285.

Describe Your Data:

Raw seismic, continuously recorded, day-long ambient noise waveforms recorded at the seismic stations in Madagascar. The list of stations at Madagascar has been obtained from Incorporated Research Institutions for Seismology (IRIS) Data Management Center - GMAP service (https://ds.iris.edu/gmap/#network=*&maxlat=-9.8499&maxlon=51.3281&minlat=-27.6739&minlon=42.1462&drawingmode=box&planet=earth). The station list has been further filtered as per the availability of data in Olugboji & Siyu (2022).

This will be downloaded from <https://doi.org/10.5281/zenodo.5753071> provided by Olugboji & Siyu (2022). This data is already preprocessed with a bandpass filter covering the standard micro-seismic noise range of 0.002 Hz to 0.4 Hz. Additionally, it included standard minimalist operations



spectrally whitened to equalize frequency contributions prior to multiplication. In the second, a multi-taper spectral estimation procedure is applied. Here, each 3-hour segment is time-normalized and multiplied by K orthogonal Slepian tapers (Thompson, 1982), selected based on the time-series length and the bandwidth of interest (typically ~ 0.05 Hz for Love waves). However, because using a large number of tapers substantially increases time and memory complexity of this algorithm, the processing is modified by selecting a reduced subset K'' of tapers drawn from the transition-region eigenspace of the Slepian tapers, such that $K'' \ll K$ (Karnik, 2022). The tapered windows are then Fourier transformed, and cross-spectra from each taper are averaged (stacked) to yield a more stable and high-fidelity frequency-domain estimate. All resulting frequency-domain NCFs are subsequently Inverse Fourier transformed back to the time domain. The following figure illustrates

The performance of the multi-taper method will be quantitatively assessed by comparing the signal-to-noise ratios (SNRs) of the resulting cross-correlation functions against those derived from conventional whitening. Additionally, for each station pair, the time-domain NCFs will be aligned and stacked according to inter-station distance to qualitatively evaluate improvements in the retrieved correlation wavefield.

To further assess the robustness of the multi-taper approach, an additional experiment will be conducted using only those noise windows recorded at least two hours after moderate-to-large earthquake events ($M_w > 5.5$) occurring during the operational period of the station network. The goal is to examine whether multi-taper processing yields enhanced correlation stability in the presence of coda. For both the general dataset and the earthquake-adjacent subset, correlation wavefields will be visualized as a function of inter-station distance and lag time to enable direct comparison of the effectiveness of multi-taper processing.

Provide References and Key Ideas

References:

- Karnik, Santhosh, Justin Romberg, and Mark A. Davenport. "Thomson's multi-taper method revisited." *IEEE Transactions on Information Theory* 68.7 (2022): 4864-4891
- Thomson, D. J. (1982). Spectrum estimation and harmonic analysis. *Proceedings of the IEEE*
- Tkalčić, H., Phạm, T.S. and Wang, S., 2020. The Earth's coda correlation wavefield: Rise of the new paradigm and recent advances. *Earth-Science Reviews*, 208, p.103285.

Key Ideas:

- Improvement of the Noise Correlation Function of Surface Love Waves
- Introducing Fast Multi-Tapers Signal Processing Technique for Cross Correlation Analysis
- Coda Correlogram Computations with Fast Multi-Taper Technique Vs Classical NCF
Signal Processing steps



How do you implement in Code

Refined Workflow for Seismic Noise Cross-Correlation and Analysis using MATLAB:

This study will implement a robust workflow in MATLAB to compute Noise Cross-Correlation Functions (NCFs) and perform advanced correlation analysis. The process is structured as follows:

1. Data Acquisition and Pre-processing:
 - Download 24-hour segments of pre-processed seismic data for all stations within the defined operating range.
 - Identify and pair stations that share coincident operational dates over the analysis period.
2. Windowing and Normalization:
 - For each valid station pair, the 24-hour signals will be divided into 3-hour overlapping windows.
 - These windows will be organized into two 3D matrices (e.g., A and B), where:
 - The X-dimension represents the time axis (samples).
 - The Y-dimension represents the day of operation.
 - The Z-dimension represents the sequential 3-hour windows within that day.
 - Each individual window segment (total $X*Y*Z$ segments per pair) will undergo temporal normalization to stabilize the noise spectrum.
3. Cross-Correlation and Stacking (Standard NCF):
 - The cross-correlation for each corresponding window pair will be computed via multiplication of the contemporary segments:
 - Apply Fast Fourier Transform (FFT) to each window segment from matrices A and B.
 - Compute the cross-spectrum by multiplying one FFT by the complex conjugate of the other.
 - Stack all cross-correlated segments to construct the final NCF.
 - Apply Inverse Fast Fourier Transform (IFFT) to return the final NCF to the time domain.
4. Alternative Method Implementation:
 - The initial steps (1-3) will be replicated for the Fast Multi-Taper (FMT) method, with Step 4 adapted to incorporate the specific spectral multiplication and stacking procedures of the Multi-Taper algorithm.
5. Coda Correlation Analysis:
 - For Coda Correlation analysis, time windows will be specifically filtered based on seismic event arrival times. Only the two windows immediately preceding and the four windows immediately following the seismic event arrival will be retained and used for correlation computation.

- Note: I will probably keep two sets, one is with seismic events and one without. Will compare the NCF wavefield for both of them. Seismicity Stack and Quiet Stack comparison.
 - Do full stack. Not just 3-6 or 6-9 hours stack separately. Choose window after earthquake and do full stack of all those windows.
- I will also need to compute NCF with 1 degree binning if possible
- Compute the slowness histogram 2D spatial direction plot for the location to introduce the data.

Results and Discussions:

Describe What you Find

This section evaluates the performance of the proposed Fast Multi-Tapered Cross-Correlation (Fast-Mspec) framework through a series of synthetic and real-data experiments. We first assess the method using a controlled spectral reconstruction test based on a synthetic noise spectrum model, allowing the intrinsic behavior and stability of the algorithm to be quantified. To examine real-world applicability, the technique is then applied to continuous ambient-noise recordings from 36 seismic stations across Madagascar Island, Africa. For each station pair, we compute the cross-correlation functions and associated cross-correlation spectra using both the conventional first-order processing and the Fast-Mspec formulation. The resulting time-domain correlation plots are compared using a signal-to-noise ratio (SNR) metric, enabling a quantitative assessment of reconstruction quality. Additionally, time-domain and frequency-domain correlation wavefields are also computed using both methods, enabling a direct comparison of their reconstruction fidelity, coherence enhancement, and overall sensitivity to ambient noise variability. Together, these analyses demonstrate not only the robustness of Fast-Mspec to complex noise conditions but also its enhanced ability to extract coherent wavefield information relative to standard cross-correlation approaches. Due to time constraints the project has not been extended to computing the Coda Correlograms. This will be addressed in future works.

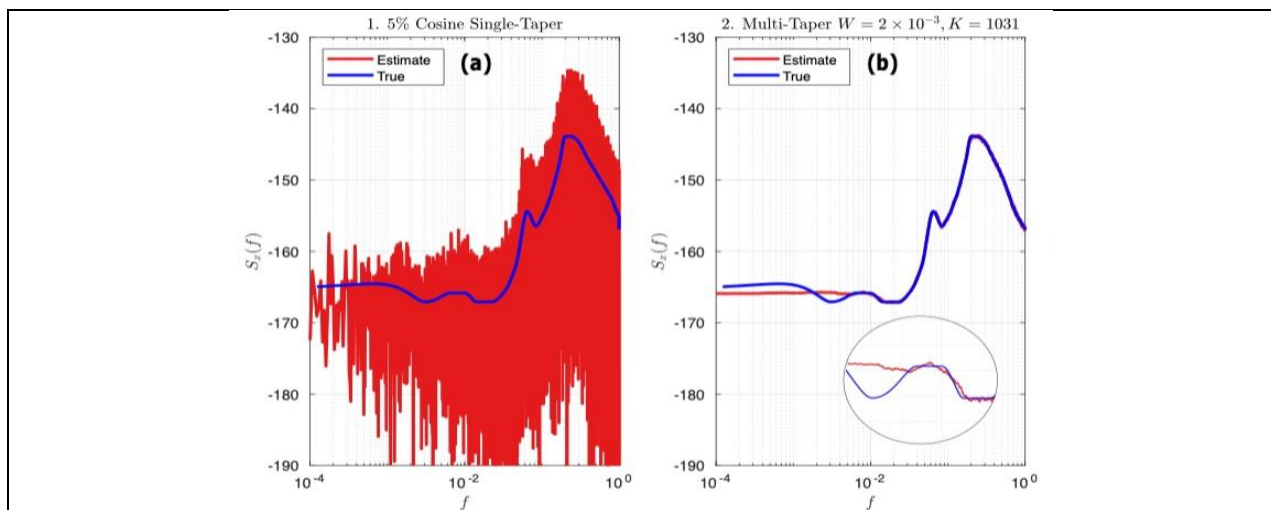
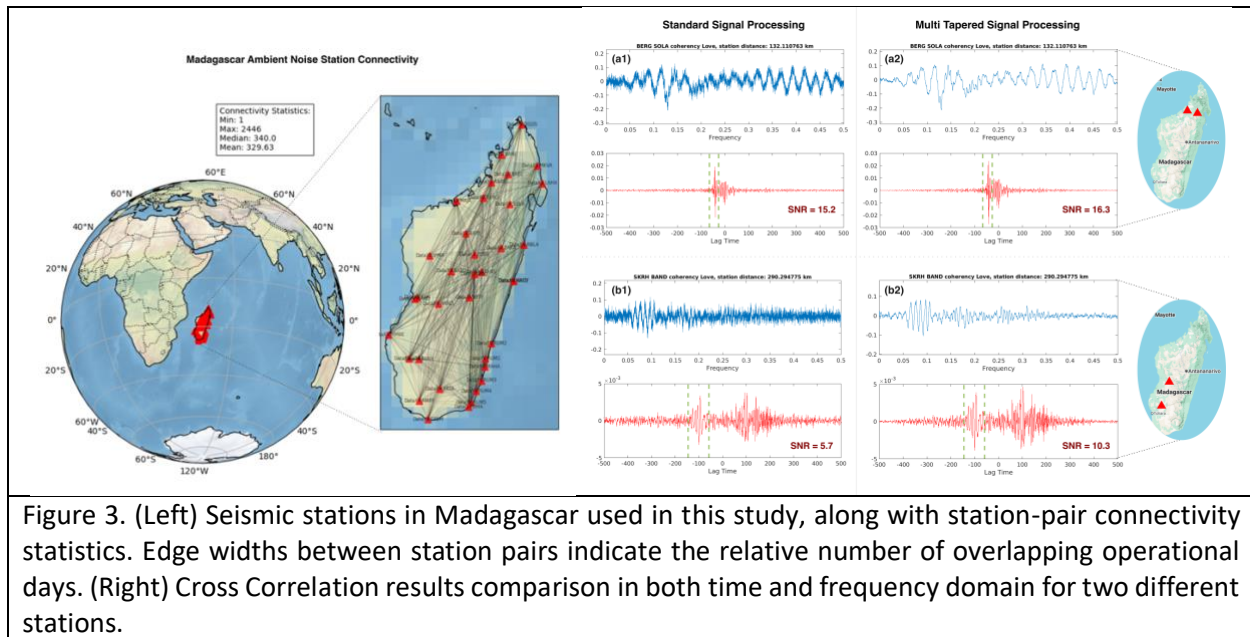


Figure 2: Spectral Bias and Variance in Power Spectral Estimation Using Multi-Taper (MT) and Single Taper (ST) Methods.

To establish a baseline, a synthetic ambient-noise waveform was first generated using the spectral characteristics of the New Low Noise Model (NLNM), following the Gaussian random process formulation outlined in Peterson (1993). Power spectral density (PSD) estimates were then computed using both the conventional single-taper method and the proposed multi-taper approach. As shown in Figure 2, the single-taper estimate in 2(a) exhibits pronounced variance, especially across frequency bands with large dynamic range, leading to visible spectral fluctuations. In contrast, the multi-taper PSD in 2(b) provides a substantially smoother and more stable reconstruction, with reduced variance and minimal bias. This initial experiment confirms the robustness of the multi-taper framework and motivates its use for subsequent cross-correlation analyses.



Next, we applied both correlation methods to the ambient seismic noise recordings from the Madagascar network. The station layout and operational connectivity are shown in Figure 3 (left). Clear differences emerged between the two approaches in the frequency-domain results: the classical single-taper method produced cross-spectra with elevated stochastic variability, particularly in bands of low noise stability, whereas the multi-taper-based FastMspec approach yielded markedly smoother and more stable spectral correlation estimates. Importantly, this improvement was achieved without introducing phase distortions or temporal delays, as no post-hoc smoothing filters were required. The time-domain cross-correlations further reflected this enhancement. Across the station pairs, the FastMspec method consistently produced higher signal-to-noise ratios compared to the traditional approach, indicating a more reliable extraction of coherent seismic energy from the ambient noise field. Figure 3 (right) demonstrates these differences for two representative station pairs.

Furthermore, the correlation-wavefield analyses computed using both techniques consistently showed improved performance for the FastMspec method showed in Figure 4. The stacked time-domain cross-correlations exhibit clearer, more coherent phase arrivals under FastMspec (Figure 4, right), whereas the classical first-order approach (Figure 4, left) retains substantial residual noise particularly for station pairs with shorter inter-station distances, where incoherent local noise tends to persist.

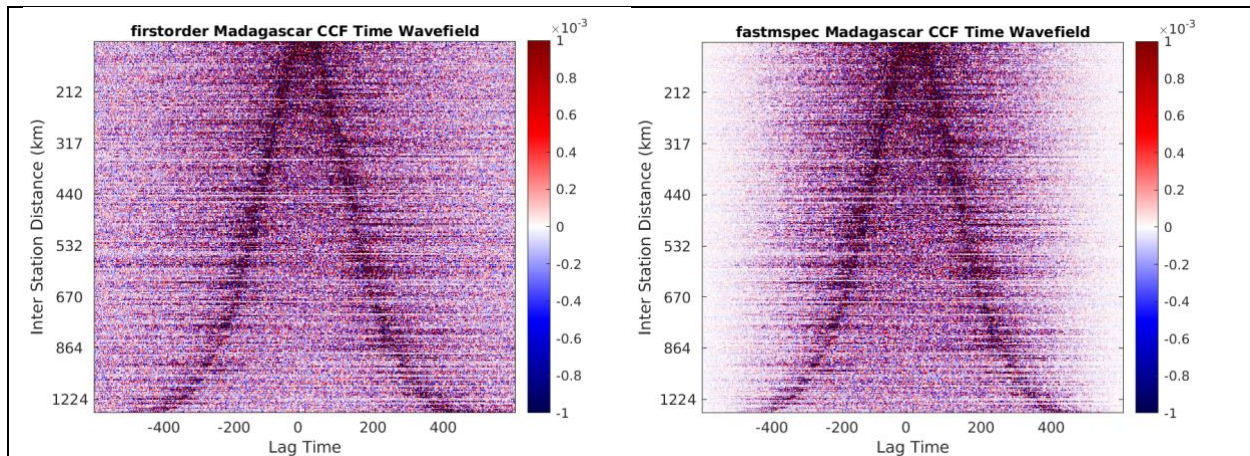


Figure 4. (Left) Short-period surface love waves across the Madagascar island obtained from classical and fast-mspec processed stacked time-domain NCFs.

The advantages of FastMspec are also evident at longer lag times (e.g., $>+400$ s), where true correlation amplitudes are expected to be weak. In these regions, the FastMspec processing produces appropriately attenuated, low-noise correlation estimates, while the conventional method continues to show fluctuations. This behavior highlights the robustness of the multi-taper formulation in suppressing stochastic variance and preserving physically meaningful correlation structure across both strong and weak signal regimes.

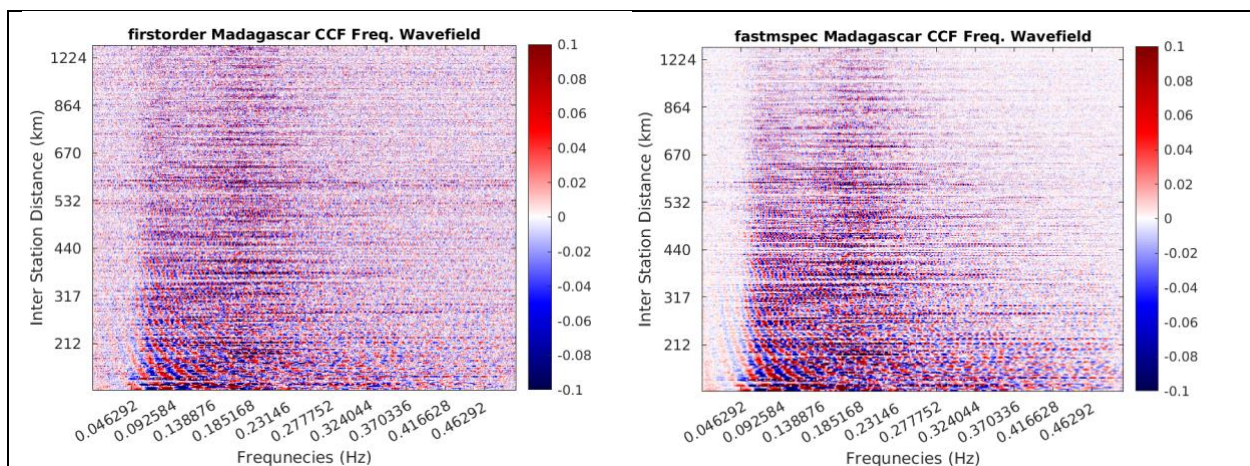
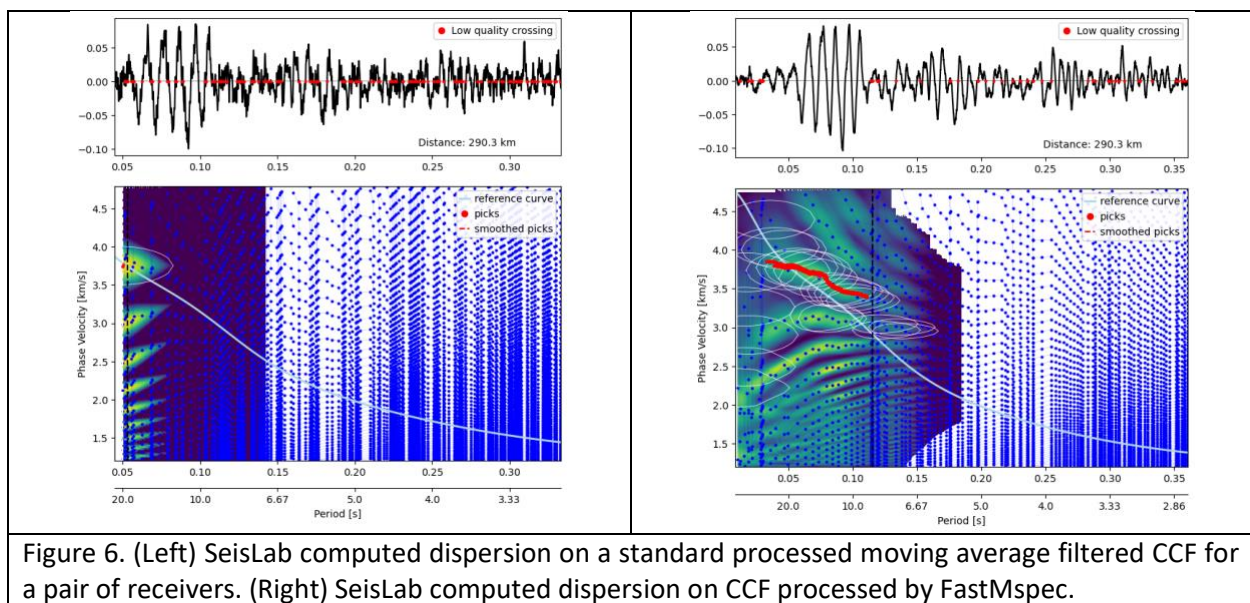


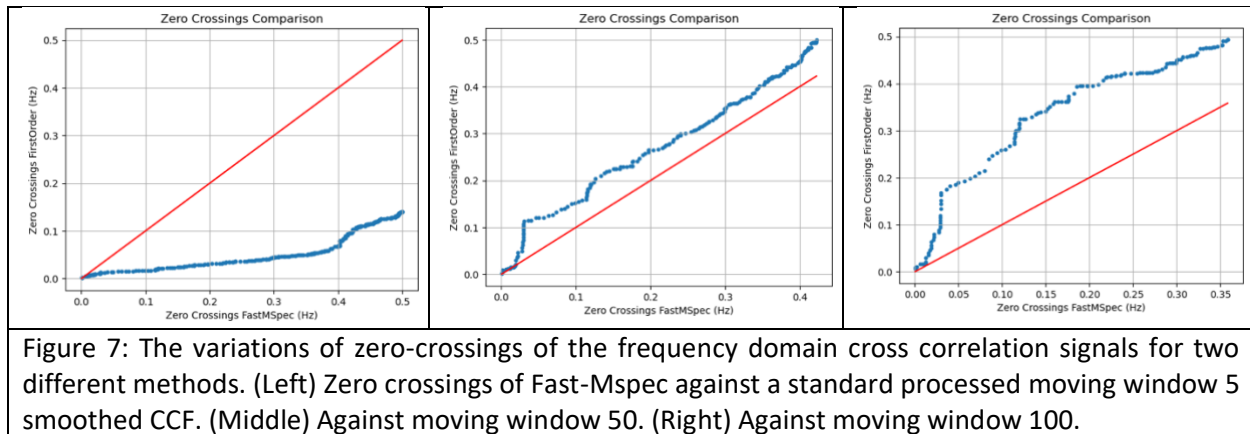
Figure 5. (Left) Short-period surface love waves across the Madagascar island obtained from classical and fast-mspec processed stacked frequency-domain NCFs.

Since the multi-taper approach improves filtering by providing a more stable and accurate spectral reconstruction, we computed the correlation wavefield in the frequency domain. The resulting cross-correlation frequency-distance diagram is analogous to a phase velocity diagram, where both velocity and group delay contribute to the appearance of dispersive trends. The frequency-distance figure reveals clear and coherent dispersive patterns characteristic of Love wave propagation across the station network. In the Fast-Mspec results (Figure 5, right), alternating red-blue bands become increasingly dense at higher frequencies, reflecting shorter wavelengths and stronger phase sensitivity to inter-station spacing. At very low frequencies, coherent energy is minimal, consistent with the limited array aperture and weak ambient-noise power in this band. Notably, high-frequency signals, which propagate more slowly, are attenuated more rapidly with distance, a behavior that Fast-Mspec captures accurately. Compared to the classical signal-processing approach (Figure 5, left), Fast-Mspec produces substantially cleaner and more stable patterns, with reduced noise contamination. This improvement demonstrates the method's effectiveness in enhancing surface-wave coherence, particularly in the mid-frequency range (0.8–2.5 Hz), where conventional techniques often suffer from low signal-to-noise ratios.



Since the goal of this study is to extract surface-wave dispersion information from ambient-noise cross correlations, we evaluated the phase-velocity retrievals obtained from Fast-Mspec-processed CCFs and those from conventionally processed CCFs. Phase velocities were estimated using the SeisLab library (Magrini et al., 2022), which computes dispersion curves from the zero crossings of the normalized cross spectrum between station pairs. The left panel of Figure 6 clearly illustrates that the high stochasticity present in the conventionally processed CCFs prevents SeisLab from identifying a stable sequence of zero crossings—resulting in poor or non-convergent phase-velocity estimates. In contrast, the Fast-Mspec CCFs, characterized by their reduced variance and more coherent spectral structure, enable consistent and robust zero-

crossing identification. Consequently, the corresponding phase-velocity retrievals show markedly improved stability and reliability.



An important consideration in phase-velocity extraction is the sensitivity of zero-crossing locations to spectral smoothing. While the amplitude at the zero crossings is irrelevant for dispersion analysis, their precise locations are critical. Excessive smoothing of the cross spectrum can artificially shift these zero crossings, thereby introducing uncertainty in the retrieved phase velocities. This effect is illustrated in Figure 6: the left panel (smoothing window = 5) preserves the zero-crossing structure reasonably well, the middle panel (window = 50) shows noticeable drift, and the right panel (window = 100) exhibits substantial shifts that would lead to unreliable dispersion measurements. This further emphasizes the advantage of Fast-Mspec, which achieves variance reduction without relying on large smoothing windows that distort zero-crossing positions.

Do results confirm expectations?

Yes, the results confirm expectations.

Did method work? Does it need to be refined?

The methods work on the synthetic and the real-world subset of ambient noise data. Although more experimentation needs to be performed and some parametric adjustment might need to be updated based on the data being processed.

References (1 or More):

- Olugboji, Tolulope, and Siyu Xue. "A Short-Period Surface-Wave Dispersion Dataset for Model Assessment of Africa's Crust: ADAMA." *Seismological Society of America* 93.3 (2022): 1943-1959.
- Karnik, Santhosh, Justin Romberg, and Mark A. Davenport. "Thomson's multitaper method revisited." *IEEE Transactions on Information Theory* 68.7 (2022): 4864-4891
- Tkalcic, H., Pham, T.S. and Wang, S., 2020. The Earth's coda correlation wavefield: Rise of the new paradigm and recent advances. *Earth-Science Reviews*, 208, p.103285.



- Peterson, Jon R. Observations and modeling of seismic background noise. No. 93-322. US Geological Survey, 1993.
- Fabrizio Magrini, Sebastian Lauro, Emanuel Kästle, Lapo Boschi, Surface-wave tomography using SeisLib: a Python package for multiscale seismic imaging, *Geophysical Journal International*, Volume 231, Issue 2, November 2022, Pages 1011–1030, <https://doi.org/10.1093/gji/ggac236>